

Zoubair MABROUK

Computer Engineer & PhD Student
AI, Software Engineering and Decision Support Systems



☎ (+216) 25 302 525 ✉ zoubair.mabrouk@fsegs.usf.tn
🌐 [LinkedIn](#) 🐙 [GitHub](#) 👤 [Portfolio](#) 📍 Sfax, Tunisia

PROFILE

Computer Engineer holding a National Engineering Degree in Software Engineering and Decision Support Systems, a Research Master's Degree in Computer Science, and currently pursuing a PhD in Computer Science. My profile combines strong expertise in software engineering, information systems, artificial intelligence, and decision support systems. I have experience designing and developing software architectures, microservices, APIs, data pipelines, and decision support solutions integrating AI approaches, particularly LLMs and RAG systems. My research background also enables me to design and evaluate experimental solutions combining data, AI models, and software prototypes.

EDUCATION

National Engineering Degree in Software Engineering and Decision Support Systems

2023 – 2026

International Institute of Technology (IIT), Tunisia

PhD in Computer Science

Research Topic: Intelligent Monitoring and Automated Anomaly Detection in the Cloud-Edge Continuum

2026 – Present

FSEGS – University of Sfax, Tunisia

Research Master's Degree in Computer Science

Research Topic: Intelligent Evolution of Decision Support Systems and Semantic Integration

2024 – 2025

FSEGS – University of Sfax, Tunisia

Bachelor's Degree in Computer Science – Embedded Systems

2020 – 2022

ISET Sfax, Tunisia

PROFESSIONAL EXPERIENCE

EDI-Solutions – *Final-Year Engineering Internship*

February – July 2026

- Designed and developed an intelligent decision support platform.
- Developed a .NET microservices architecture with an API Gateway.
- Designed and integrated APIs for the different platform services.
- Implemented automatic generation of MDX queries from natural language using RAG approaches.
- Integrated dynamic dashboards and a multidimensional ETL pipeline.
- Containerized services using Docker and integrated local AI components.
- **Technologies:** C#, ASP.NET Core, SQL Server, SSAS, MDX, Docker, LangChain, FAISS, Ollama.

ERITECH Groups – *Engineering Internship*

July – August 2025

- Adapted, integrated, and customized open-source home automation solutions.

Confiserie Triki le Moulin – Sfax – *Final-Year Bachelor's Internship*

February – June 2022

- Designed an Industry 4.0-based monitoring platform.
- Developed a mobile application for real-time monitoring and dashboarding.
- Integrated industrial data using OPC UA.
- **Technologies:** JavaScript, Java (Android), Firebase, Raspberry Pi 3, Linux, OPC UA, Industry 4.0.

SKILLS

Software Engineering & Development: C#, ASP.NET Core, Java/Spring Boot, React/TypeScript, REST APIs, Microservices, Software Architecture, Git/GitHub, CI/CD.

AI & Machine Learning: Python, PyTorch, TensorFlow, NLP, Transformers, LLMs, RAG, Sentence Transformers, Embeddings, FAISS, Scikit-Learn, Ollama.

Data & Decision Support Systems: SQL Server, PostgreSQL, MongoDB, Data Warehouse, ETL, OLAP, SSIS, SSAS, Power BI, MDX, Data Integration, Schema Matching.

Infrastructure & Tools: Docker, Linux, Git/GitHub, CI/CD, API Gateway, LangChain.

BPM & Process Management: BPMN, Process Mining, Sustainable development of Business Process.

RESEARCH & SCIENTIFIC PUBLICATION

- Said Taktak, **Zoubair Mabrouk**, Slim Kallel, Ahmed Hadj Kacem: *Managing Model Evolution from NoSQL Data Lakes to Decision Support Systems*, Proceedings of the 14th International Conference on Model and Data Engineering (MEDI), Cairo, Egypt, November 2–4, pages 61–80, 2025. [DOI](#)

CERTIFICATIONS

- NVIDIA – Getting Started with Deep Learning (2026).
- Big Data & Machine Learning with Apache Spark (2024).
- Python Object-Oriented Programming (2024).